

Rituraj Singh

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CAREER OBJECTIVE

AI/ML engineer specializing in CV, NLP, and LLM infrastructure, focused on building and deploying scalable, production-grade AI systems and intelligent automation pipelines.

EDUCATION

Dayananda Sagar Academy of Technology & Management <i>B.E., Computer Science and Engineering (Artificial Intelligence) — CGPA: 8.95, GPA: 3.56</i>	Bangalore, India 2023 – 2027
Indian Institute of Technology, Madras (IITM) <i>B.S., Data Science and Applications</i>	Chennai, India 2024 – 2027
St. Karen's High School <i>X: 94% (2021) XII: 77.2% (2023)</i>	Patna, India

INTERNSHIP & WORK EXPERIENCE

Spaceborn <i>AI/ML & LLM Intern</i> - Developed and optimized the schema and validator for reducing hallucination, the Q-LoRA pipeline, multi-drone routing, and a FastAPI service. - Built an A* and D* Lite path planning algorithm system on a 4D temporal map for MVP2 with real-time obstacle avoidance. - Achieved success in MVP1 development and evaluation; promoted to the core AI and LLM engineering team.	Bangalore, India March – July 2026
Microsoft — Azure AI <i>Workshop Intern</i> - Developed an OCR system for handwritten digits with Scikit-learn SVM, achieving ~98.9% accuracy through effective data preprocessing and model training. - Built a real-time digit recognition interface with Streamlit, enabling practical applications in bank check processing and document digitization.	Bangalore, India May – June 2025
Google Summer of Code 2025 <i>OpenCV and OpenMS Submissions</i>	

RESEARCH

Style-Consistent Motion Transfer from Walking to Arbitrary Actions <i>Intl. Conference on Intelligent Computing and Sustainable Technologies</i>	Ref: SCRS/ICST 2026/PC/263 ICST 2026
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PROJECTS

OpenCV LLM Tokenizer <i>[Code]</i> - Developed OpenCV's first native LLM Tokenizer framework as part of Google Summer of Code, implementing BPE, WordPiece, Unigram, SentencePiece, TikToken, and Hugging Face integration within the DNN module. - Optimized for high performance with SIMD acceleration, CUDA-based GPU batch processing, and multi-threading, achieving significant speedup for multilingual text tokenization. - Designed a unified API with Python & C++ bindings, supporting adaptive multilingual tokenization, benchmarking tools, and seamless integration with multimodal AI pipelines in OpenCV.
CogInfera <i>[Code]</i> - Developed an advanced RAG-based system for complex document understanding, implementing hierarchical ingestion pipelines with PDF parsing, semantic chunking, and embedding generation for high-quality contextual retrieval. - Engineered a hybrid retrieval framework combining dense vector search (FAISS) and sparse retrieval (BM25), enhanced with reranking and agent-driven query planning for efficient multi-hop information extraction. - Designed a modular, production-grade architecture with cognitive agents (Query Planner, Retrieval Controller) and multi-pass Self-RAG reasoning.
HVAC Optimization Sys <i>[Code]</i> - Designed an AI-powered HVAC optimization system integrating YOLOv5 (occupancy detection), OpenCV, PyTorch, and the OpenWeatherMap API for real-time adaptive climate control. - Combined a lightweight Transformer-based regression model for climate prediction with a PID controller and rule-based fallback system, balancing energy efficiency and thermal comfort. - Built an interactive Streamlit dashboard with real-time monitoring, historical analytics, simulation mode, and CSV-based data logging.
FinAgent — ML Investment Strategist <i>[Code]</i> Engineered an AI-powered strategist that transforms financial profiles and risk tolerance into optimized portfolios with rationale, using Streamlit, FinBERT, and Gemma.

- Implemented modern portfolio optimization (MPT, Ledoit-Wolf), sentiment analysis, and rebalancing recommendations, integrating real-time market and fundamental data.
- Simplified complex financial analysis for non-experts through interactive Plotly visualizations and robust, cached data pipelines.

DeepSeq — Advanced Protein Analysis Tool | [\[Code\]](#)

- Built an AI-powered tool that predicts protein 3D structures, functions, and stability using ESMFold, embeddings, and ML-based property analysis.
- Simplified protein characterization with interactive Py3Dmol visualizations, residue-level insights, and functional annotation.
- Improved research efficiency through real-time analysis, stability assessment, and automated pipelines with validation and caching.

TECHNICAL SKILLS

Programming: C, C++, Python, Java, Julia, R, JavaScript, TS, Rust, Node.js, Bun, MySQL, PostgreSQL, MongoDB, VectorDB

Libraries/Frameworks/APIs: NextJS, ReactJS, Flask, TensorFlow, PyTorch, Scikit-Learn, Keras, NumPy, Pandas, Ollama, OpenCV, Hugging Face, FastAPI, LangChain/LangGraph, CUDA, Pybind11/Nanobind, GitHub Actions, Docker, Grafana, Prometheus, PySpark, Kafka, MLflow, Airflow, TensorBoard, llama.cpp, ONNX, REST, gRPC

Non-Technical: Critical Thinking, Analytical Skills, Learning Agility, Communication, Leadership, Teamwork

CERTIFICATIONS

- Anthropic Academy — Model Context Protocol: Advanced Topics
- LinkedIn — High Performance PySpark
- Docker — Docker Foundations Professional Certificate
- ISRO-IIRS — RS & GIS Applications in Natural Resource Management
- DeepLearning.AI — Natural Language Processing in TensorFlow
- Georgia Tech — Supply Chain Principles
- NVIDIA — AI Infrastructure and Operations Fundamentals
- NVIDIA — AI for All: From GenAI to Basics
- Postman — API Fundamentals Student Expert

EXTRACURRICULARS

Winner — LUMINAX 2k25 Hackathon

- Developed *Surge*: a real-time environmental monitoring system tracking AQI and WQI with real-time data visualization, interactive maps, and AI-powered predictions.

Build with AI — Google Developer Groups GGITS

- Developed *YT: Summae & Synopsis*; gained recognition from fellow participants and the judging panel.

AICTE Problem Statement: AI Hospital Staffing Solution

- Developed an intelligent staffing system with containerization, robust pipelines, and monitoring.

Core Group Member

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| • Maker Space — Coding and Innovation Club, DSATM | Oct 2024 – Present |
| • Wayanad House — 1 of 12 Houses of IIT Madras | Sep 2024 – Present |
| • GDG IIT Madras — Active Member | Jan 2025 – Present |
| • GirlScript Summer of Code 2025 — Mentor | Jul 2025 – Oct 2025 |

RELEVANT COURSEWORK

Core CS: Data Structures and Algorithms, Discrete Mathematics, Theory of Computation, Computer Organization & Architecture, Operating Systems, Introduction to Unix/Linux, Compilers, Introduction to Cyber Security

Math & Statistics: Linear Algebra and ODE, Analysis & Multivariate Calculus, Eigenvalues/Eigenvectors & Game Theory, Probability for Computer Science, Statistics for Computer Science

AI/ML: Machine Learning (Advanced), Probabilistic Machine Learning, Deep Learning for Computer Vision, Data Analytics with R

Other: Remote Sensing & GIS (IIRS), Internet of Things & Networking, Introduction to Electronics & Communication